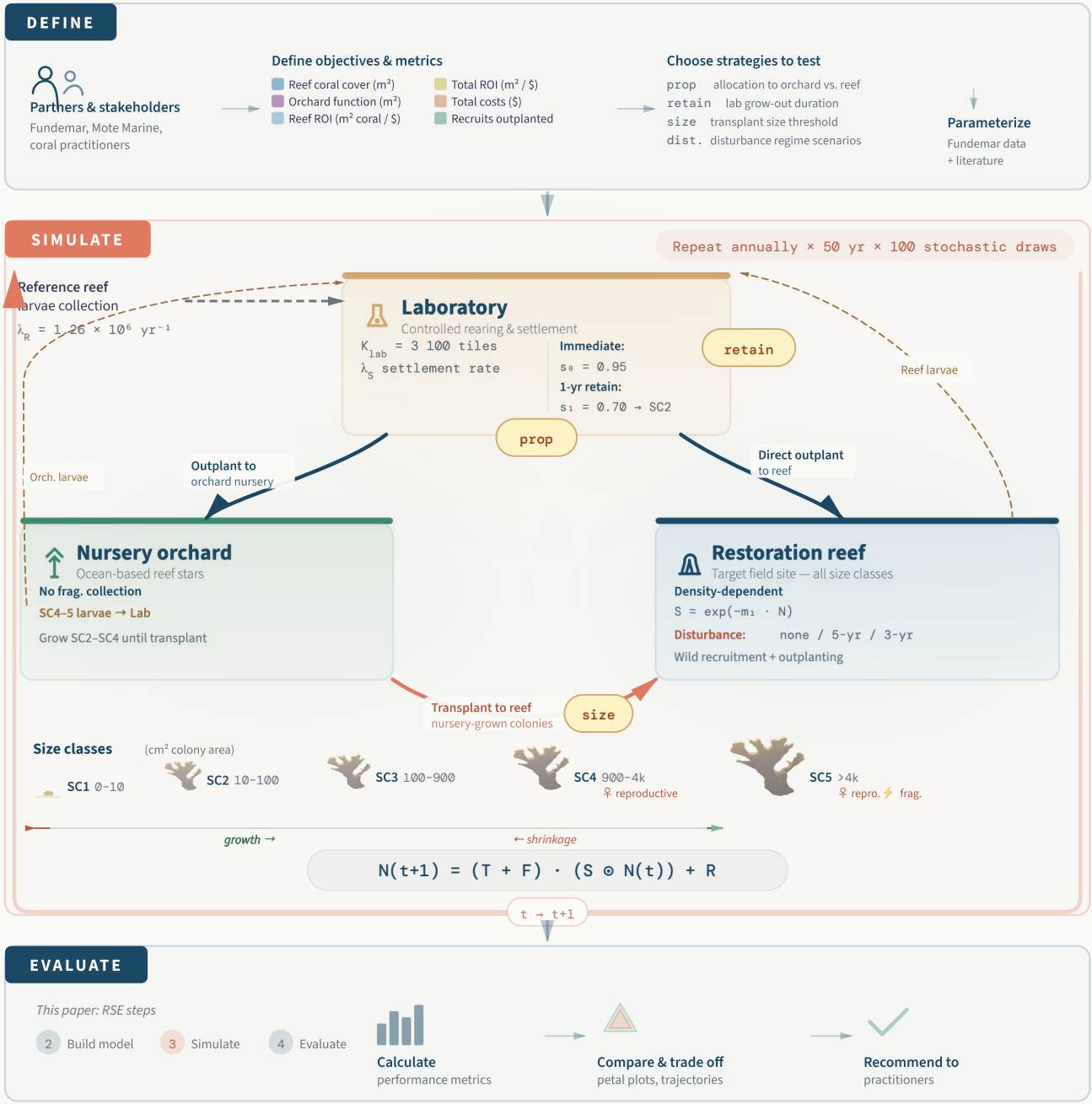
DEFINE
Partners & stakeholders
Fundemar, Mote Marine, coral practitioners
Define objectives & metrics
Reef coral cover (m²)
Orchard function (m²)
Reef ROI (m² coral / $)
Total ROI (m² / $)
Total costs ($)
Recruits outplanted
Choose strategies to test
prop allocation to orchard vs. reef
retain lab grow-out duration
size transplant size threshold
dist. disturbance regime scenarios
Parameterize
Fundemar data + literature
SIMULATE
Repeat annually × 50 yr × 100 stochastic draws
Reference reef larvae collection
λR = 1.26 × 10⁶ yr⁻¹
Laboratory
Controlled rearing & settlement
Klab = 3 100 tiles
λS settlement rate
Immediate:
s0 = 0.95
1-yr retain:
s1 = 0.70 → SC2
retain
prop
Reef larvae
Orch. larvae
Outplant to orchard nursery
Direct outplant to reef
Nursery orchard
Ocean-based reef stars
No frag. collection
SC4–5 larvae → Lab
Grow SC2–SC4 until transplant
Restoration reef
Target field site — all size classes
Density-dependent
S = exp(−m1 · N)
Disturbance: none / 5-yr / 3-yr
Wild recruitment + outplanting
Transplant to reef
nursery-grown colonies
size
Size classes
(cm² colony area)
SC1 0–10
SC2 10–100
SC3 100–900
SC4 900–4k
♀ reproductive
SC5 >4k
♀ repro. ⚡ frag.
growth →
← shrinkage
N(t+1) = (T + F) · (S ⊙ N(t)) + R
t → t+1
EVALUATE
This paper: RSE steps
2 Build model
3 Simulate
4 Evaluate
Calculate
performance metrics
Compare & trade off
petal plots, trajectories
Recommend to
practitioners
Ecological flow
Management decision
External input
Larvae feedback
decision
S survival
T transition
F frag.
R recruitment
⊙ element-wise